

Title:

E-Commerce Sales Data Analysis Using MySQL Analytical Functions.

Problem Statement:

A multinational retail company operates an ecommerce platform across multiple cities, segments, and product categories. The management team needs a deeper understanding of their sales performance, customer ordering patterns, and product-level profitability to make data-driven decisions. They require insights to identify top revenue-generating categories, determine which cities drive the highest order volumes, track sales trends over time, and distinguish between high-performing and loss-making products using structured SQL analysis.

Objective:

To analyze ecommerce sales performance and profitability using MySQL's core SQL functionalities, including aggregation, grouping, filtering, and sorting.

Dataset Overview

The dataset contains transactional sales records for products sold through the ecommerce platform. Each row represents a single order line item with details on the customer, product, pricing, and profitability. The key columns include:

Column	Description
order_id	Unique identifier for each order
order_date	Date the order was placed
ship_date	Date the order was shipped
ship_mode	Shipping mode selected for the order
customer_id	Unique identifier for the customer
customer_name	Name of the customer
segment	Customer segment (e.g., Consumer, Corporate, Home Office)
city	City where the order was placed

Column	Description
state	State where the order was placed
country	Country where order was placed
region	Region of the order (e.g., East, West, Central, South)
product_id	Unique identifier for the product
category	Category of the product (e.g., Technology, Furniture, Office Supplies)
sub_category	Subcategory of the product
product_name	Name of the product
sales	Total sales amount for the order
quantity	Quantity of items ordered
discount	Discount applied to the order
profit	Profit generated from the order
month	Month the order was placed
year	Year the order was placed

Task:

1. Find the categories generating the highest sales revenue.
2. Identify the top cities in terms of number of orders.
3. Find out the monthly sales trend.
4. Find the top 10 most profitable products.
5. Find out the loss-making products.